

Why a Steepest-Descent Line Search at $k = 0$ is Consistent with *Minimization of expensive functions in the presence of forbidden regions*

1 The degeneracy at $k = 0$

`ffjm2` minimizes $f(x) = \frac{1}{2} \|F(x)\|^2$, where $F = (f_1, \dots, f_m)^\top$ is the residual vector. At external iteration k it keeps one symmetric matrix H_i^k per residual component f_i , approximating $\nabla^2 f_i(x^k)$, and solves

$$M_k(s) = \sigma \|s\|^2 + \frac{1}{2} \sum_i q_i(s)^2, \quad q_i(s) = f_i(x^k) + \nabla f_i(x^k)^\top s + \frac{1}{2} s^\top H_i^k s, \quad (1)$$

accepting or rejecting $x^k + s$ by the ratio between actual and predicted reduction, and growing/shrinking σ accordingly (this is the regularization scheme of Algorithm 4.1 of [1], implemented in `ffjm2` with an empirical persistent- σ variant — the paper resets $\sigma \leftarrow 0$ at the start of every iteration and never shrinks it, but persisting it between iterations and shrinking it on acceptance performed better in practice).

At $k = 0$, every $H_i^0 = 0$ (no secant pair has been observed yet), so (1) collapses to

$$M_0(s) = \sigma \|s\|^2 + \frac{1}{2} \sum_i (f_i(x^0) + \nabla f_i(x^0)^\top s)^2, \quad (2)$$

a single damped Gauss–Newton subproblem. Two things are worth noting about this particular instance of the subproblem:

1. Unlike every subsequent iteration, $\sigma = \sigma_0$ here is a fixed constant, not yet informed by any accept/reject test on *this* problem. Whether it makes (2) behave like Gauss–Newton ($\sigma \rightarrow 0$) or like scaled steepest descent ($\sigma \rightarrow \infty$, where the minimizer of (2) is asymptotically parallel to $-\nabla f(x^0)$) is essentially a guess about the local curvature scale of f at x^0 that the method has had no chance to calibrate yet.
2. There is, by construction, no curvature information at all to justify solving a quartic (rather than a quadratic) model here: the quartic term $\frac{1}{2} s^\top H_i^k s$ is identically zero for every i .

So the very first step is the one iteration where the method’s own machinery (quasi-Newton curvature, calibrated regularization) has nothing to work with.

2 A self-contained argument: the closed form of M_0

Rather than appealing to a different algorithm, the cleanest justification comes directly out of (2) itself, which is an unconstrained, strictly convex quadratic in s and can be solved in closed form. Write $J := F'(x^0)$ for the Jacobian of F at x^0 (the matrix whose i -th row is $\nabla f_i(x^0)^\top$), so that $g := \nabla f(x^0) = J^\top F(x^0)$, and $A := J^\top J$ (symmetric positive semidefinite). Then

$$\nabla_s M_0(s) = (2\sigma I + A) s + g = 0 \implies s(\sigma) = -(A + 2\sigma I)^{-1} g.$$

Lemma A (descent for every $\sigma > 0$). Since $A + 2\sigma I \succ 0$ for $\sigma > 0$, its inverse is symmetric positive definite, hence

$$g^\top s(\sigma) = -g^\top (A + 2\sigma I)^{-1} g < 0 \quad \text{whenever } g \neq 0.$$

So $s(\sigma)$ is a genuine descent direction for f for every $\sigma > 0$, not just asymptotically — the σ -loop never has a “wrong-direction” problem at $k = 0$; what it can have is a badly-sized step, which is what the accept/reject test and the growth of σ are actually correcting for.

Proposition B (alignment with $-g$ as $\sigma \rightarrow \infty$). Factor 2σ out of the inverse:

$$s(\sigma) = -(A + 2\sigma I)^{-1}g = -\frac{1}{2\sigma} \left(I + \frac{A}{2\sigma} \right)^{-1} g.$$

As $\sigma \rightarrow \infty$, $A/(2\sigma) \rightarrow 0$, and matrix inversion is continuous at the identity, so $(I + A/(2\sigma))^{-1} \rightarrow I$. Hence

$$s(\sigma) = -\frac{1}{2\sigma} g + o\left(\frac{1}{\sigma}\right) \implies \frac{s(\sigma)}{\|s(\sigma)\|} \longrightarrow -\frac{g}{\|g\|} \quad \text{as } \sigma \rightarrow \infty. \quad (3)$$

In words, for σ large enough, the solution of the degenerate $k = 0$ subproblem (2) already is, to leading order, a steepest-descent step of length $\approx 1/2\sigma$. The two are asymptotically the same computation. This is a direct consequence of (2)’s own closed form, entirely internal to Algorithm 4.1.

In practice, the rejection ladder $\sigma_0, \sigma_{j+1} = \gamma\sigma_j$ ($\gamma > 1$ the growth factor) tries $s(\sigma_0), s(\sigma_1), \dots$ until one is accepted. By Proposition B, once $\sigma_j \gg \|A\|/2$ the successive trials are already essentially parallel to $-g$ and shrink by the fixed factor $1/\gamma$, so once it reaches that scale, the ladder is already behaving like a backtracking line search along $-g$. Make a linesearch along $-g$ is a good option, because the next model will use a $\sigma_1 = \sigma_0/10$. If, the first step need to be small, σ_0 will be big and the next model can be affected.

References

- [1] F. A. Fortunato and J. M. Martínez, *Minimization of expensive functions in the presence of forbidden regions*, unpublished manuscript, 2026. (`_research/ffjm2.pdf` in this repository.)